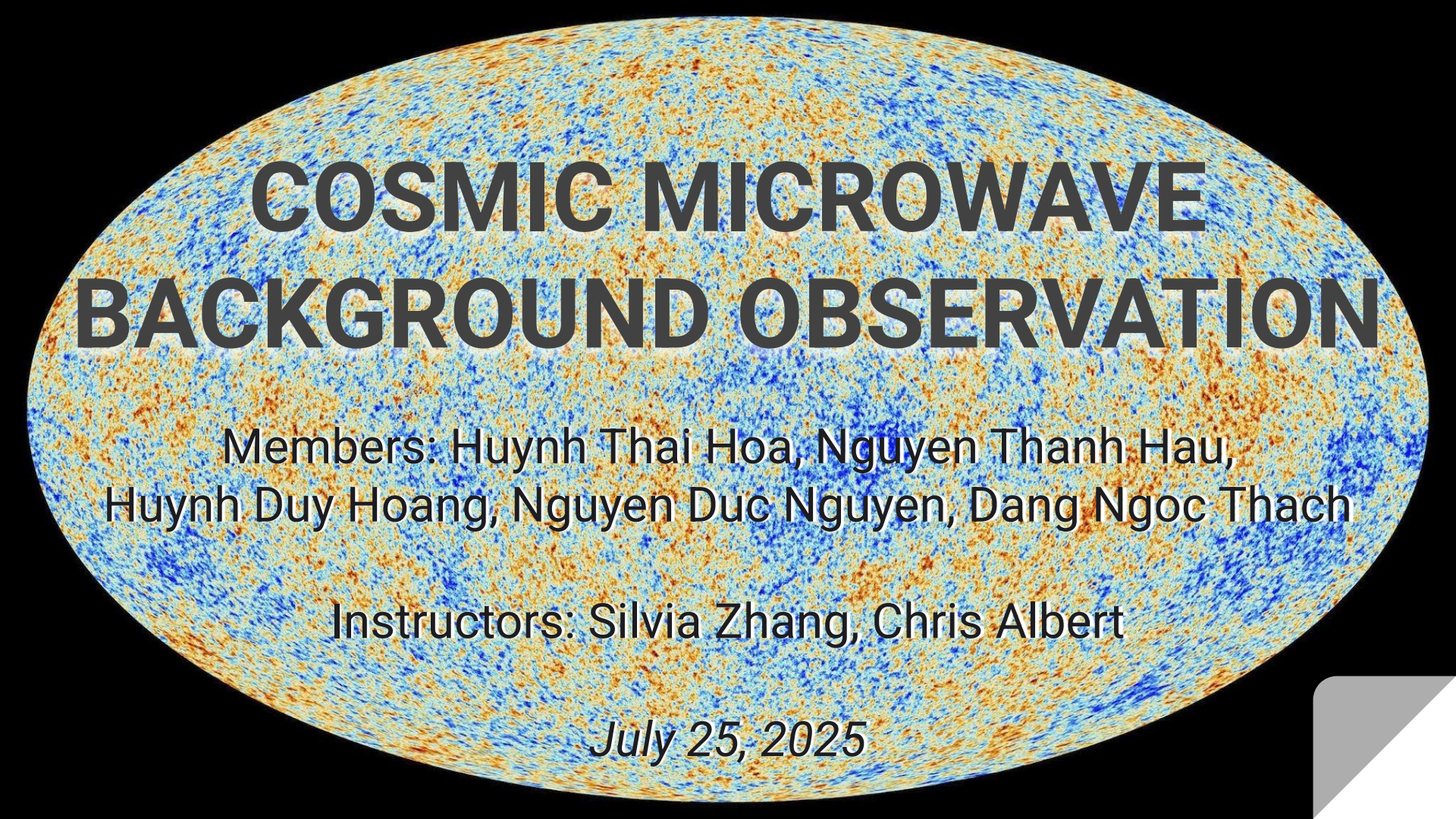


The background of the slide is a full-frame image of the Cosmic Microwave Background (CMB) fluctuation map. It shows a complex pattern of blue and orange/yellow speckles and swirls, representing temperature variations across the sky. The map is centered and fills most of the frame.

# **COSMIC MICROWAVE BACKGROUND OBSERVATION**

Members: Huynh Thai Hoa, Nguyen Thanh Hau,  
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Instructors: Silvia Zhang, Chris Albert

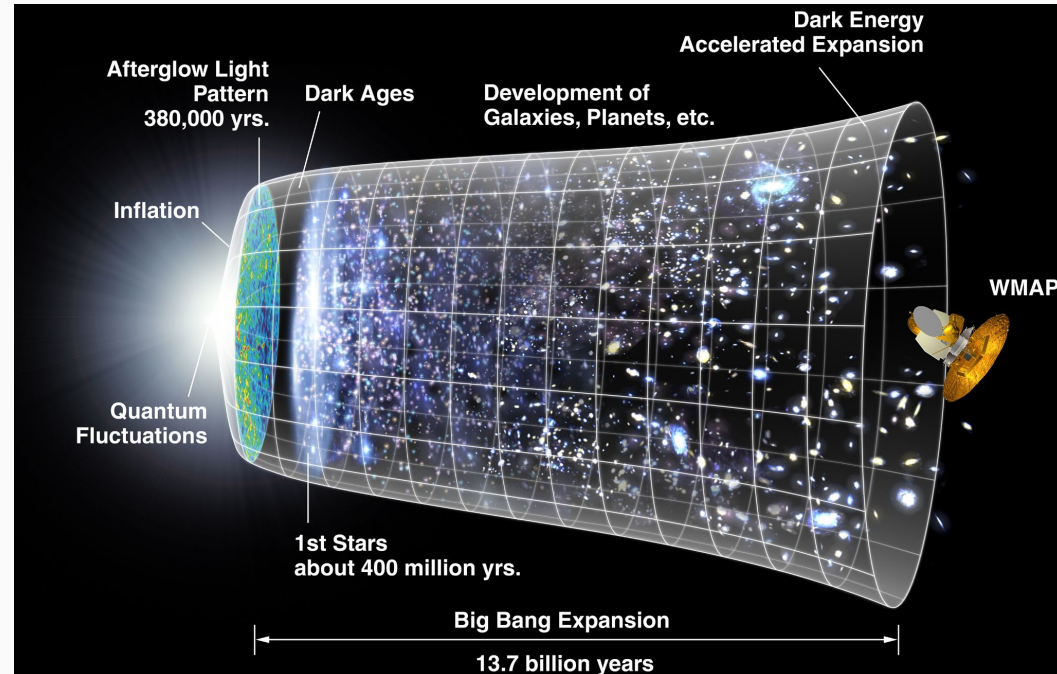
*July 25, 2025*

# Table of contents

1. Introduction
2. Equipments setup
3. Data acquisition
4. Data analysis
5. Result and discussion

# 1. Introduction

- After Big Bang, the universe consisted of hot, dense plasma.
- Photons were scattering off the free electrons.
- When it cooled to 3,000 K, e- and p combined to neutral H atoms (recombinations) → photon could travel freely through space → CMB release.

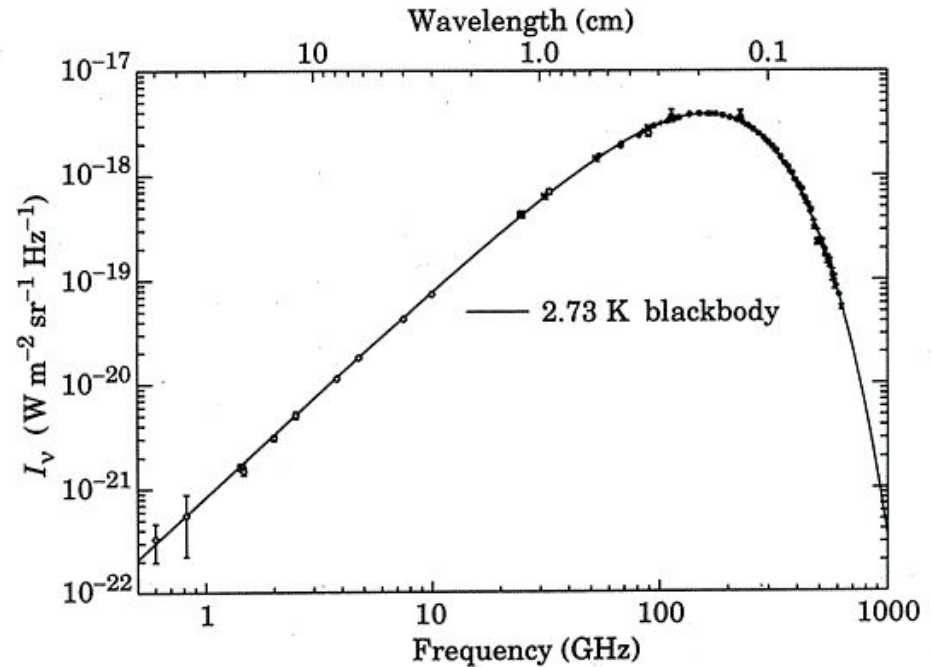


CMB is the afterglow of the big bang



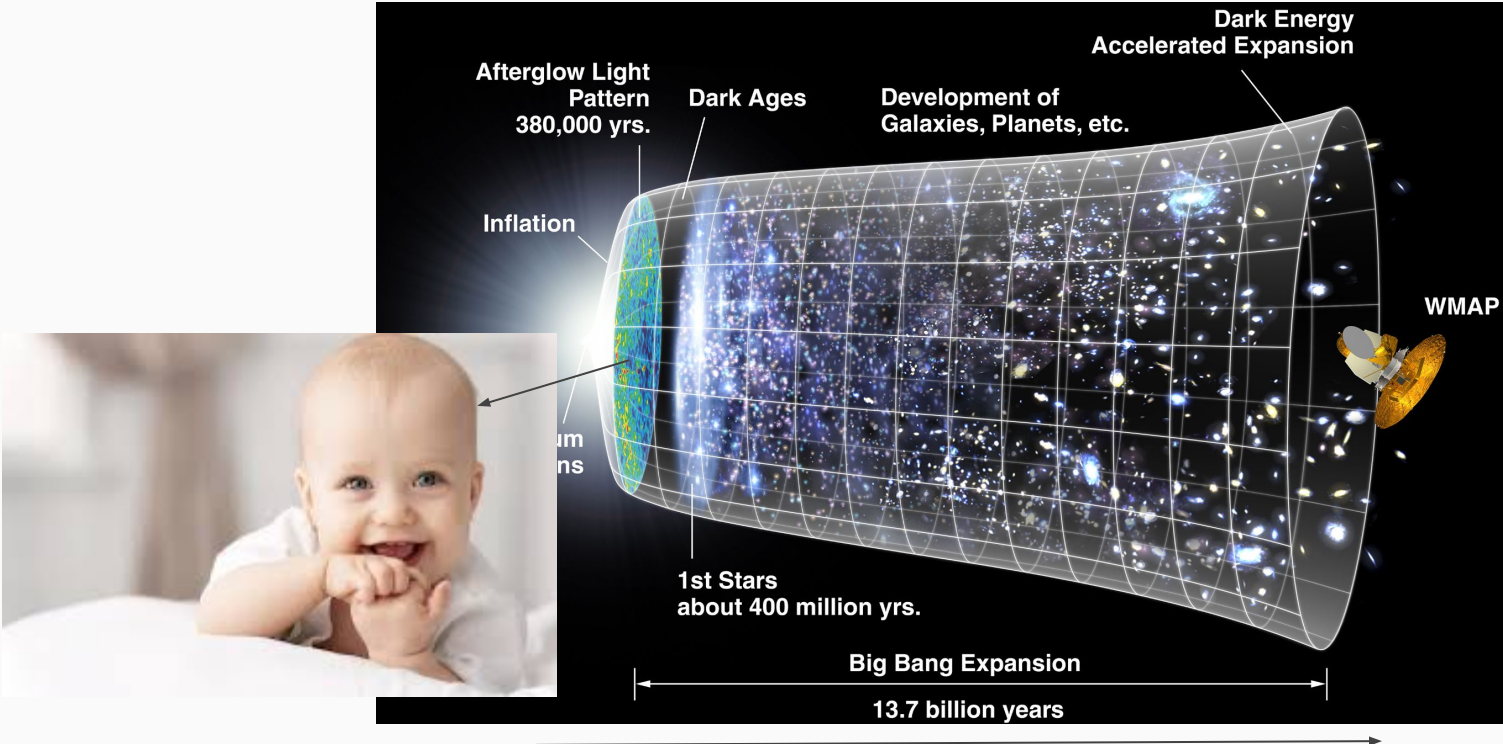
# 1. Introduction

- Formed 380,000 years after Big Bang in epoch of recombination.
- After travelling throughout the universe, it is redshifted by  $z \approx 1100$
- $T = 2.725$  K.
- Almost a perfect blackbody spectrum.



The CMB has some unique properties that make it very interesting

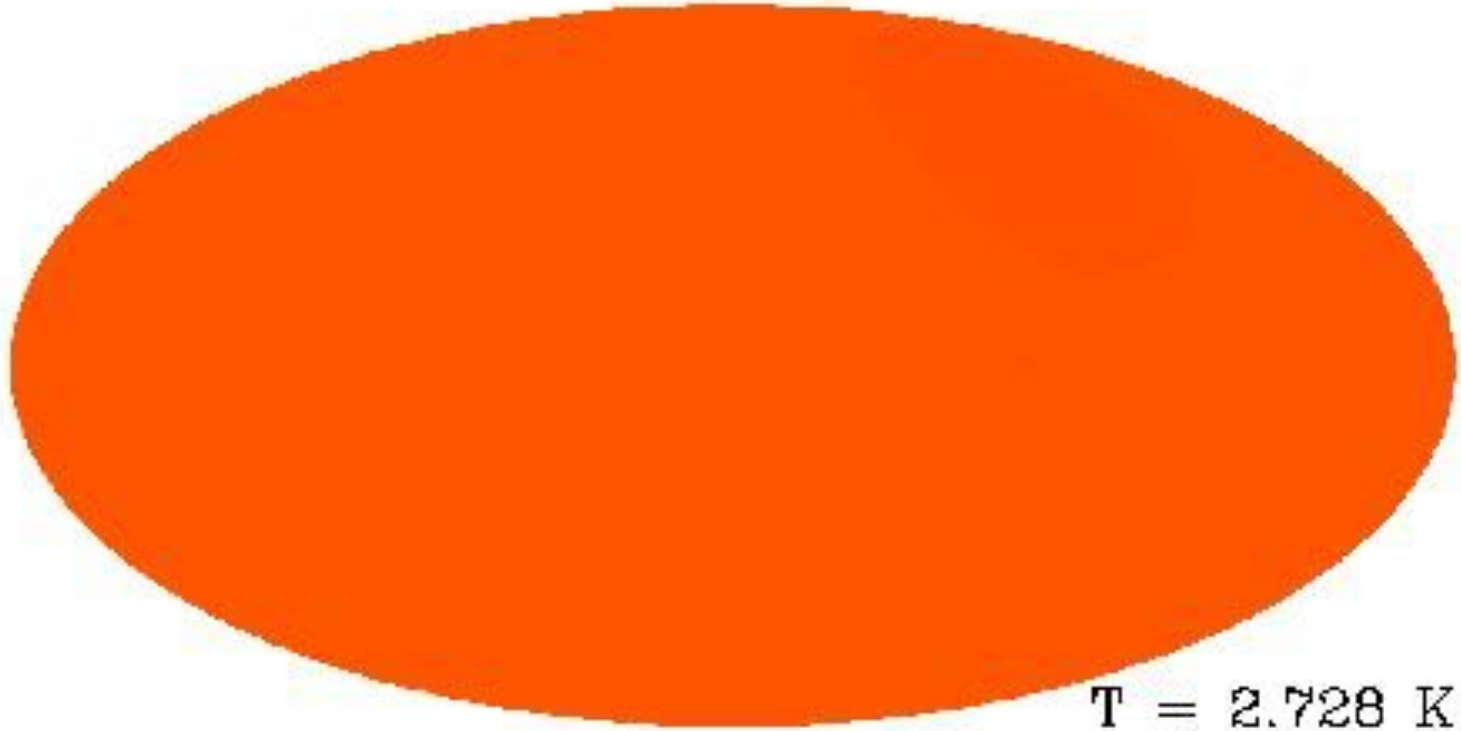
# 1. Introduction



Imagine in human timeline

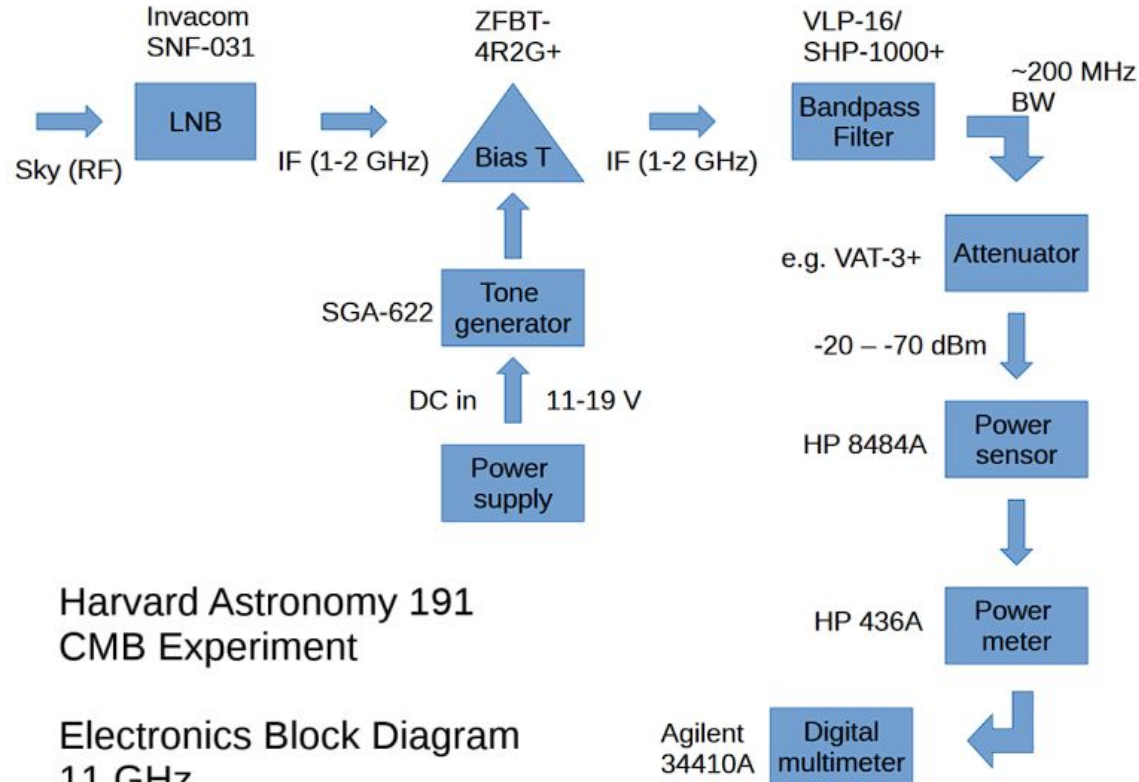
Basically it is the baby picture of the universe

# 1. Introduction



The CMB is very uniform and is basically the same at all angles

## 2. Equipments setup: Original Setup



The original idea of our setup if everything went to plan. Easy right?

## 2. Equipment setup: Power meter



Our power meter wasn't working.



## 2. Equipment setup: De\_FUSE



We replaced multiple fuses. Still not working.

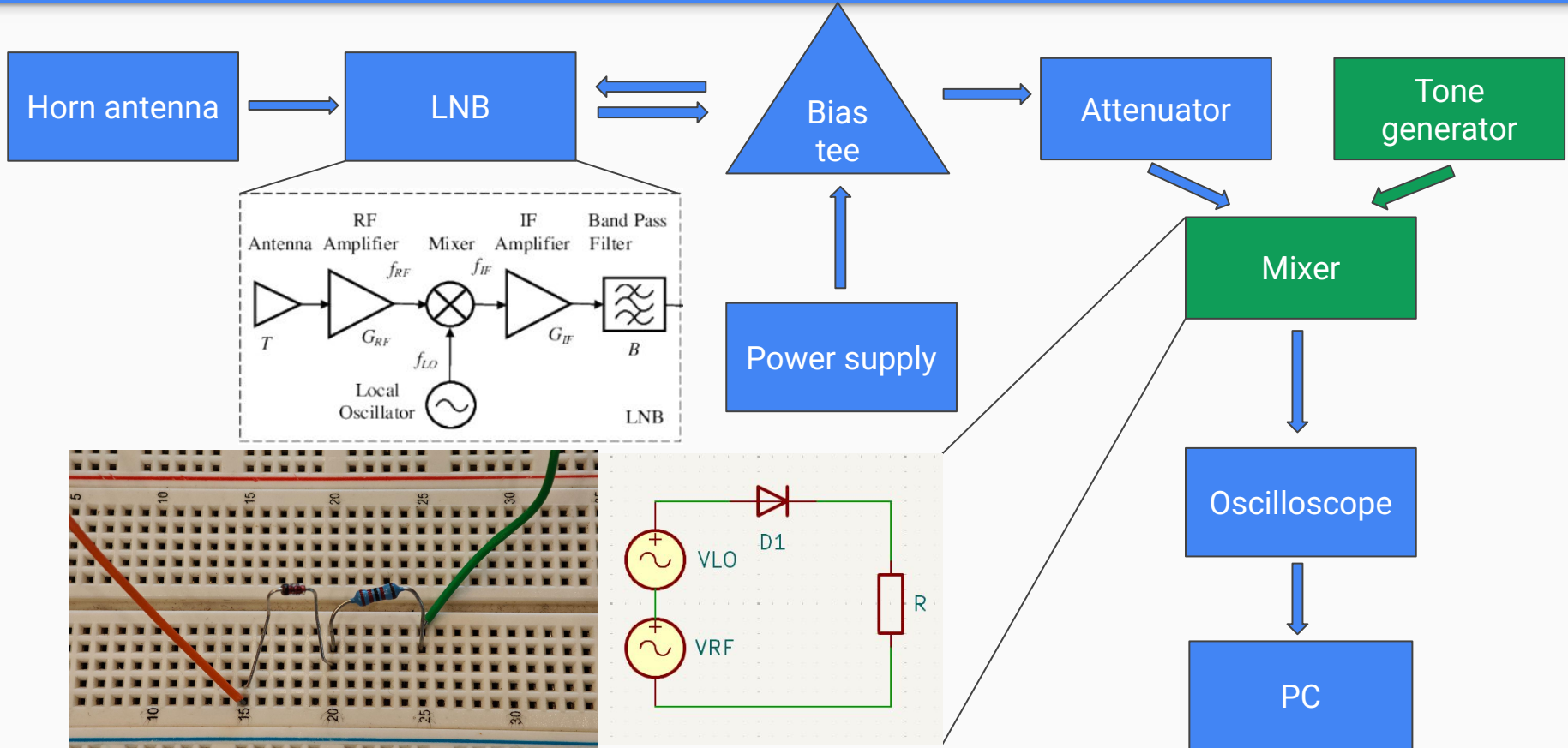
## 2. Equipment Setup

- Our solution: Replace the power meter with an oscilloscope and we do the filtering with our PC, but it only reads up to 200 MHz.



Replace power meter with an oscilloscope

## 2. Equipments Setup



Added a DIY mixer to lower the frequency of the signal, so that our oscilloscope can read it

## 2. Equipments Setup



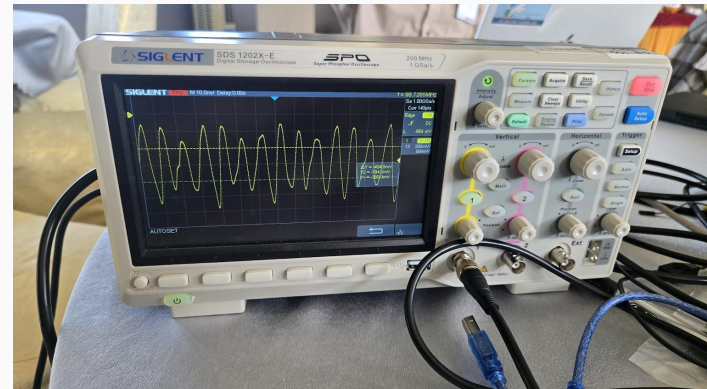


## 2. Equipments Setup



### 3. Data acquisition

- Step 1: Insert a mixing signal from the tone generator.
- Step 2: Point the conical horn antenna at the zenith ( $\theta = 0$ ).
- Step 3: Change attenuation until the signal shows up on the scope.
- Step 4: Run data acquisition code.
- Step 5: Increase zenith angle.
- Step 6: Go back to step 4 until you complete the sweeping.



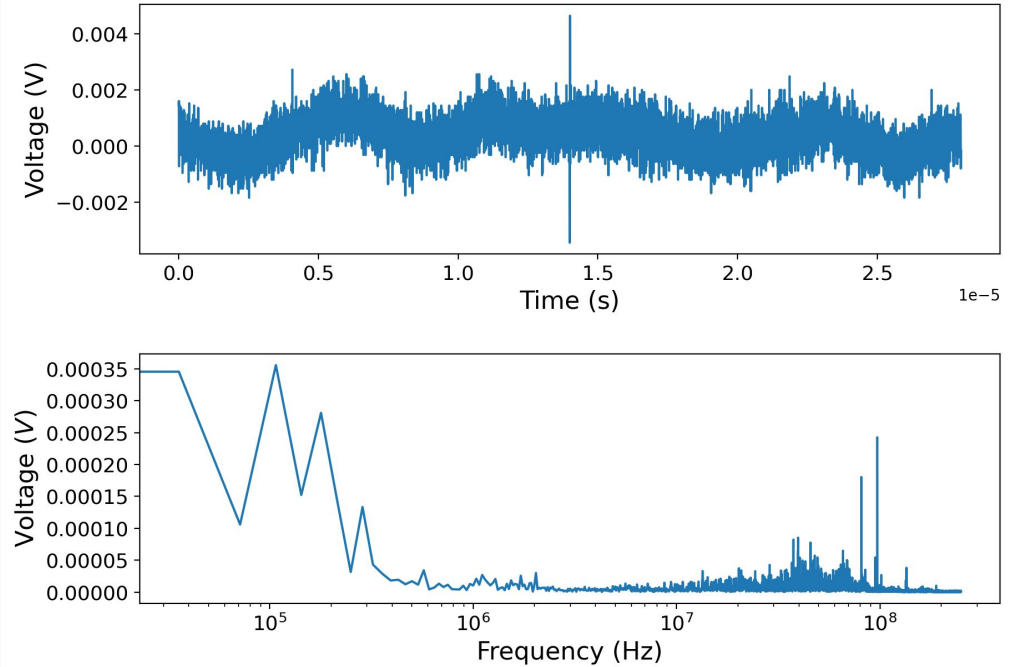
### 3. Data acquisition

- Step 7: Place a black-painted metallic tray in front of the horn to capture 300K load spectrum.
- Step 8: Point the horn to another tray immersed in liquid nitrogen to capture 77K load spectrum.



## 4. Data analysis

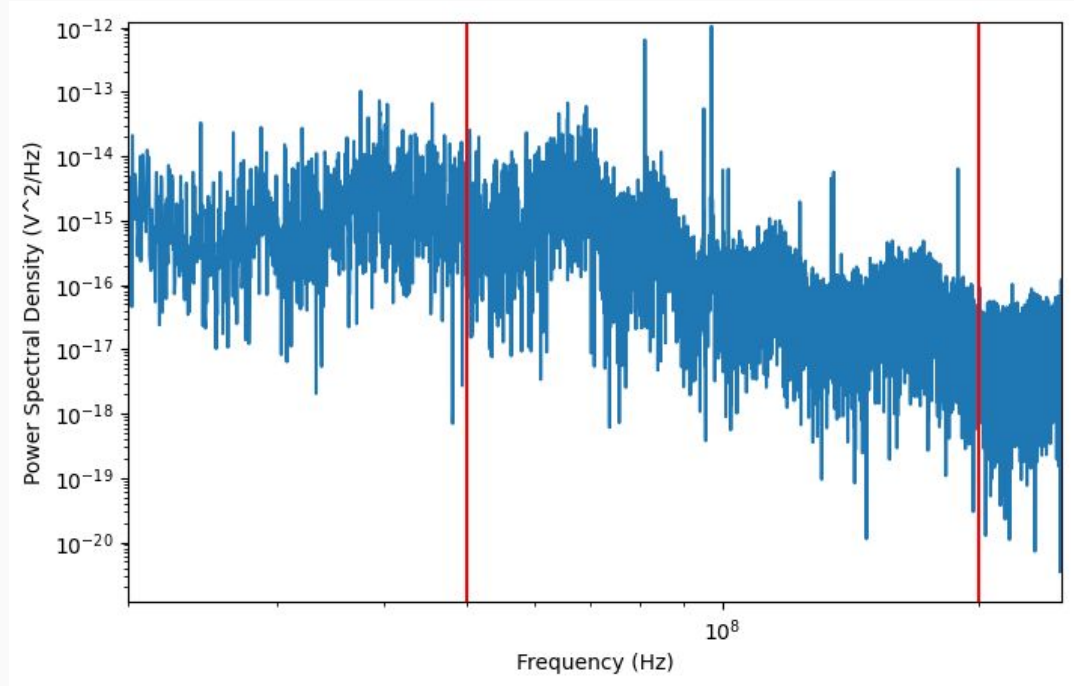
- We saved the time-series voltage data and its FFT as CSV files.





## 4. Data analysis

- Create PSD graph by applying the Wiener–Khinchin theorem.
- Applying Parseval's theorem, find the power of a signal in our targeted frequency range by integrating the PSD.
- Red lines shows the integration window.



## 4. Data analysis

- The power picked up by the horn (in units of temperature) of anything the horn is pointing at

$$T_{obs} = \frac{P_{obs}}{G} - T_{rec} \quad (1)$$

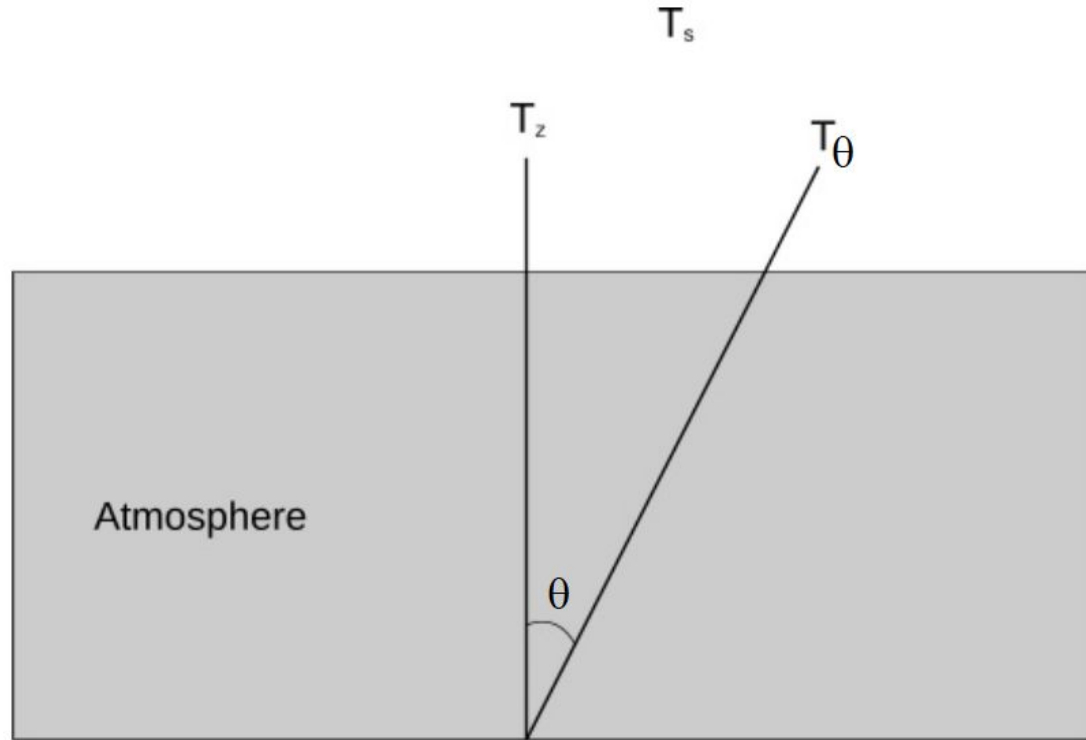
where  $G$  is the overall gain of the system, and  $T_{rec}$  is the receiver noise.

- $T_{rec}$  and  $G$  can be calculated as

$$T_{rec} = \frac{T_w P_c - T_c P_w}{P_w - P_c} \quad (2), \quad G = \frac{P_w - P_c}{T_w - T_c} \quad (3)$$

where  $T_w = 300$  K;  $T_c = 77$  K;  $P_c$  is the measured power at 77 K;  $P_w$  is the measured power at 300 K.

## 4. Data analysis



The CMB is constant everywhere, but atmospheric noise is not

## 4. Data analysis

- The sky temperature at zenith angle  $\theta$  can be expressed as

$$T_{\theta} = T_s + T_A \sec \theta \quad (4)$$

where  $T_s$  is the sky temperature without atmospheric noise;  $T_A$  is the atmospheric noise at the zenith;  $\sec \theta$  is the linear noise-to-distance coefficient - air mass. (4)

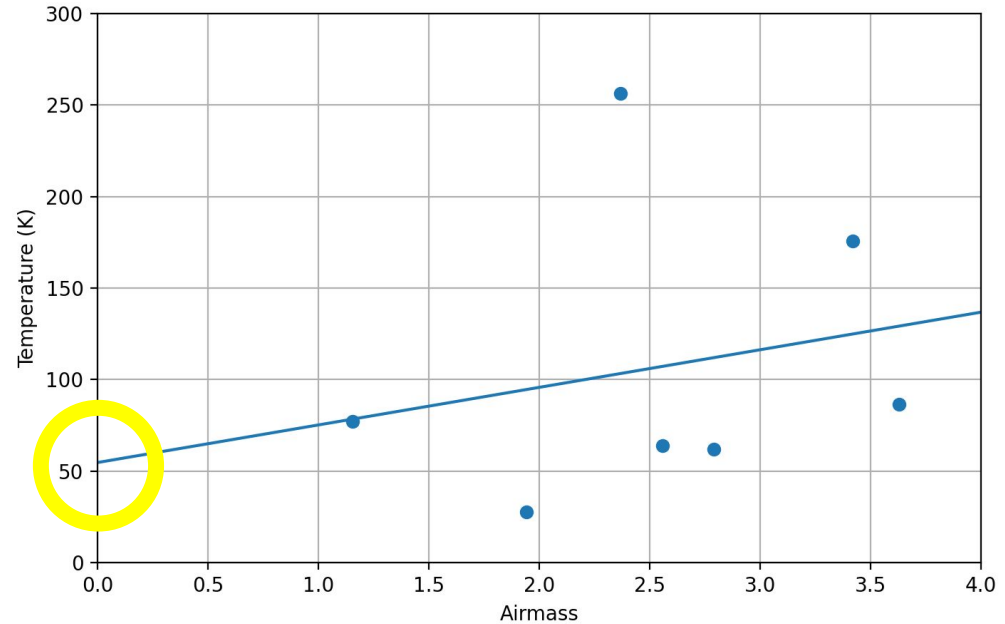
- We can make a graph of  $T_{\theta}$  vs air mass and fit the linear model above into it. The best fit  $T_s$  will be our  $T_{\text{CMB}}$ .

$T_s$



## 5. Result and discussion

- Y-intercept: 54.66K  
Standard error: 43.98
- Large error reflects limited sample size and gain fluctuations.



The CMB is the leftover signal after subtracting all the noise

## 5. Result and discussion

- Sources of error
  - Noise from the equipments (DIY mixer, etc. ).
  - Radiation from the ground.
  - High humidity
  - Intermittent cloud coverage
  - Uncertainty in angle due to mount ~only 1.5 degree precision



## 5. Result and discussion

To understand our system we need:

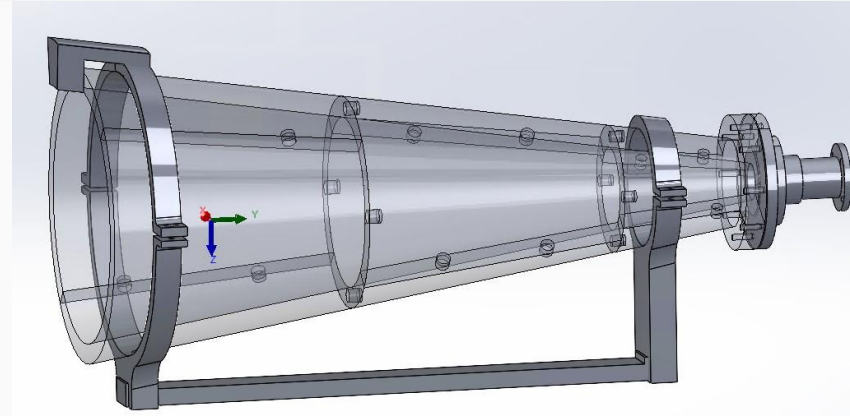
$$SNR = \frac{P_{CMB}}{P_{Noise}} \approx 1.161$$

where

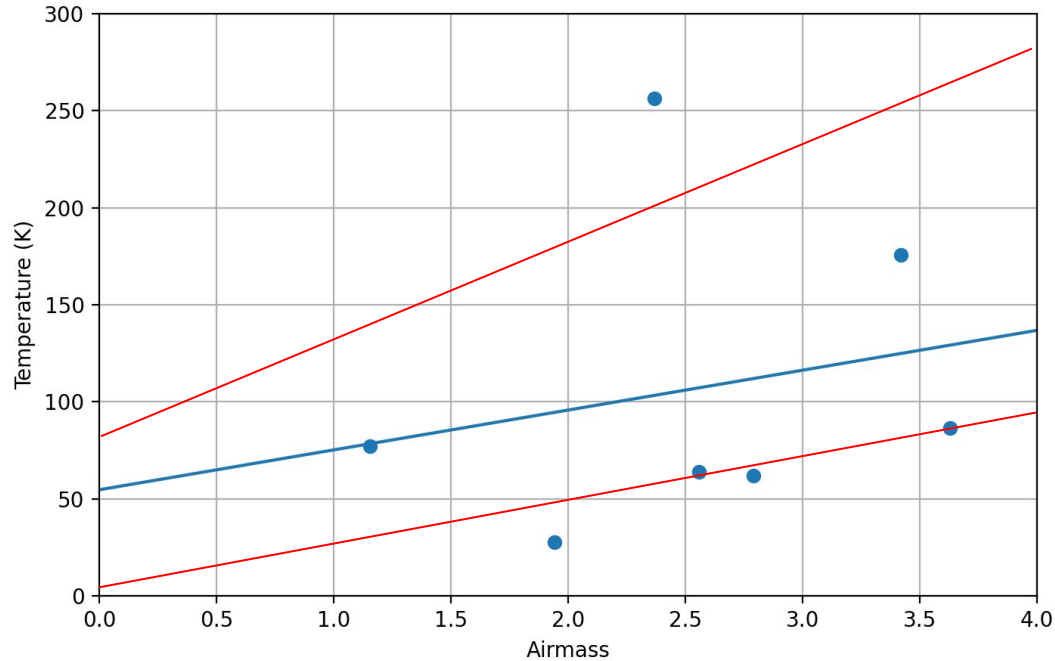
$$P_{Noise} = Bk_B T_{rec}$$

and

$$P_{CMB} = \int_{11.2GHz}^{11.35GHz} B_{CMB} A_{\Omega} d\nu$$



## 5. Result and discussion



Y-intercept: 54.66K  
Standard error: 43.98

Clouds

Clear sky

We can interpret our results as an effort to characterize all types of noise

# References

1. The cosmic microwave background. (1986, June 2). Retrieved from <https://webhome.phy.duke.edu/~kolena/cmb.htm>
2. How big was the source of the cosmic microwave background radiation 13.7 billion years in the past? - National Radio Astronomy Observatory. (2021, March 31). Retrieved from <https://public.nrao.edu/ask/how-big-was-the-source-of-the-cosmic-microwave-background-radiation-13-7-billion-years-in-the-past/>
3. Tyson, T., Barnett, B., & Hillbrand, S. (2019, April). *Measurement of the cosmic microwave background radiation at 19 GHz.*



## Special thanks



Cảm ơn

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Takk

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Gracias

Danke

Obrigado

THANK YOU

Ευχαριστώ

감사합니다

תודה

Salamat

භට්ඨකුඤ

Teşekkür ederim

Tack

ありがとう

Asante

धन्यवाद

Спасибо

شكراً

Ngiyabonga

谢谢

धन्यवाद